Regression and Trend Analysis

Meaning: Regression and trend analysis help identify patterns in time-dependent data and understand how values change over time.

Regression Analysis: Regression analysis identifies the relationship between time and observed values by fitting a suitable mathematical model. It helps estimate values and predict future outcomes based on historical data.

Trend Analysis: Trend analysis identifies the long-term direction of data, such as an increasing, decreasing, or stable pattern.

Importance: Together, these techniques help detect patterns, understand past behaviour, and forecast future values.

Example: They can be used to analyse monthly sales and predict future demand, helping businesses plan their inventory.

Seasonal ARIMA and Non-Seasonal ARIMA

Non-Seasonal ARIMA: Non-seasonal ARIMA models time-series data without explicitly modelling recurring seasonal patterns. It uses autoregressive, differencing, and moving-average terms. It is represented as ARIMA (p,d,q) and is suitable for data without strong seasonality.

Seasonal ARIMA (SARIMA): Seasonal ARIMA models data containing recurring seasonal patterns. It includes both non-seasonal and seasonal components and is represented as SARIMA (p,d,q)(P,D,Q,s), where (s) denotes the seasonal period.

Key Difference: Non-seasonal ARIMA focuses on general time-series patterns, whereas seasonal ARIMA additionally captures regular seasonal variations.

Examples:

- Non-seasonal ARIMA: Forecasting daily demand without a recurring seasonal pattern.

- Seasonal ARIMA: Forecasting ice-cream sales that increase every summer.

ARIMA Modelling Approach for Real-World Forecasting

Meaning: ARIMA (AutoRegressive Integrated Moving Average) is a statistical model used to analyse historical time-series data and forecast future values.

Step 1: Data Collection: Collect historical monthly sales data from a shop to identify past sales patterns.

Step 2: Stationarity Check: Examine whether the data has a relatively stable mean and variance over time. Apply differencing if necessary to make the series stationary.

Step 3: Model Identification: Use Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots to help select suitable model orders.

Step 4: Model Fitting: Estimate the parameters of the selected ARIMA model using the historical data.

Step 5: Model Evaluation: Examine the residuals and measure forecast accuracy using suitable error metrics.

Step 6: Forecasting: Use the fitted model to predict future monthly sales.

Application: The forecasts help the shop plan inventory, manage resources, and make informed business decisions.

Similarity Search in Time-Dependent Sequences

Meaning: Similarity search identifies time-series sequences that closely resemble a given sequence.

Comparison of Patterns: It compares historical and current sequences using distance measures such as Euclidean distance and Dynamic Time Warping (DTW).

Handling Time Differences: DTW can compare sequences with similar patterns even when their events occur at different speeds or times.

Pattern Recognition: Similarity search helps identify recurring trends, recognise repeated behaviour, and detect unusual observations or anomalies.

Forecasting Applications: Similar historical sequences can provide useful information for estimating future behaviour.

Example: A business can compare current sales patterns with previous months to identify similar periods and support sales forecasting.

Residual Component in Time-Series Decomposition

Meaning: The residual component represents the variation remaining after the trend and seasonal components have been separated from a time series.

Irregular Fluctuations: It captures random variations and other changes that are not explained by the trend or seasonal patterns.

Assessment of Decomposition: Residual analysis helps determine whether the decomposition has successfully captured the major patterns in the data.

Interpretation: Residuals randomly distributed around zero generally suggest that the main patterns have been captured effectively. However, systematic patterns in the residuals may indicate that important information remains unexplained.

Importance: Studying residuals helps identify model limitations, improve forecasting methods, and assess the reliability of the time-series analysis